
ST3101: 2. Exercises on solving linear equations**1. Solving for a vector in a system of equations**

For equations

$$\mathbf{Ax} = \mathbf{b}$$

where

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$

find x and y and draw the (column) vectors

- $\mathbf{a}_1$,
- $\mathbf{a}_2$,
- $\mathbf{b}$, and
- $x\mathbf{a}_1, y\mathbf{a}_2$.

You can reason/guess your way to the right values, no formal solutions needed.

(a)

$$\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 2 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

(c)

$$\begin{bmatrix} 1 & 2 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

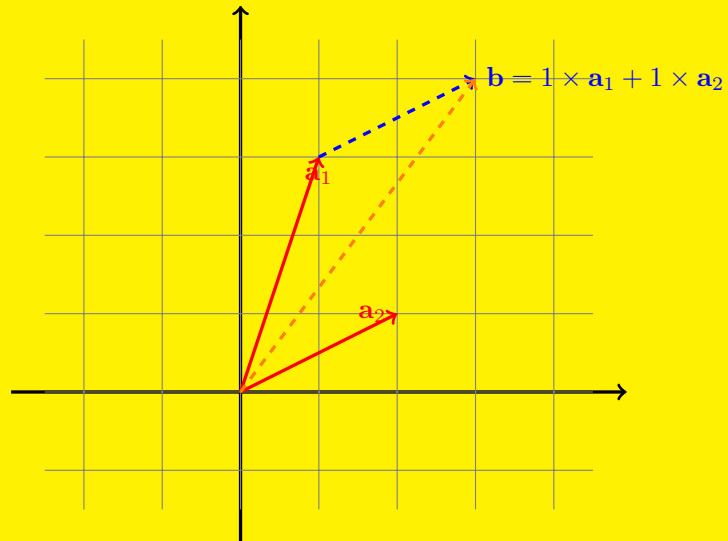
(d)

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

(e)

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

(a) We see immediately that if you add $\mathbf{a}_1$ and $\mathbf{a}_2$ you get $\mathbf{b}$. Hence we need just $x = 1$ of column 1 and $y = 1$ of column 2

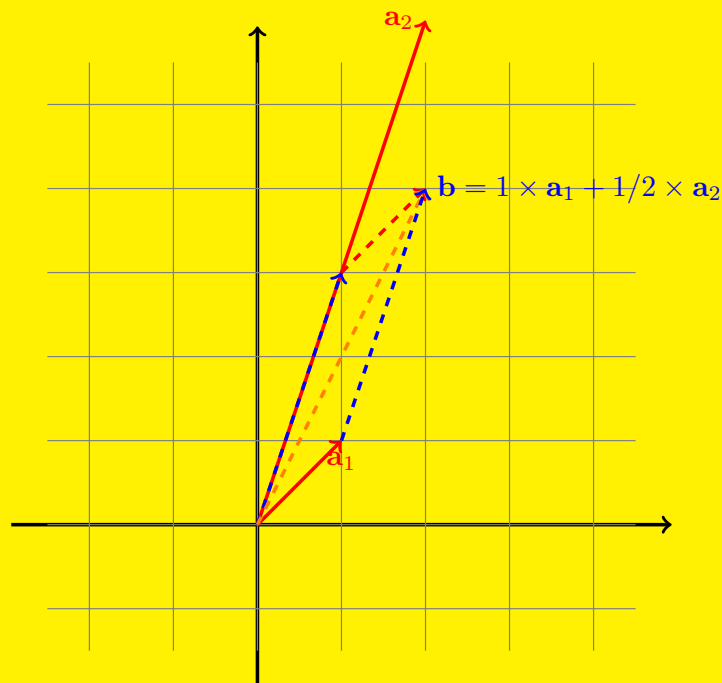


(b) The first column is all ones so we need to add 1 and 3 respectively to get $\mathbf{b}$. If we divide $\mathbf{a}_2$ by 2 we get exactly 1 and 3. Take $x = 1$ of column 1 and $y = 1/2$ of column 2, and you get

$$1 + (1/2)(2) = 2$$

and

$$1 + (1/2)(3) = 4$$



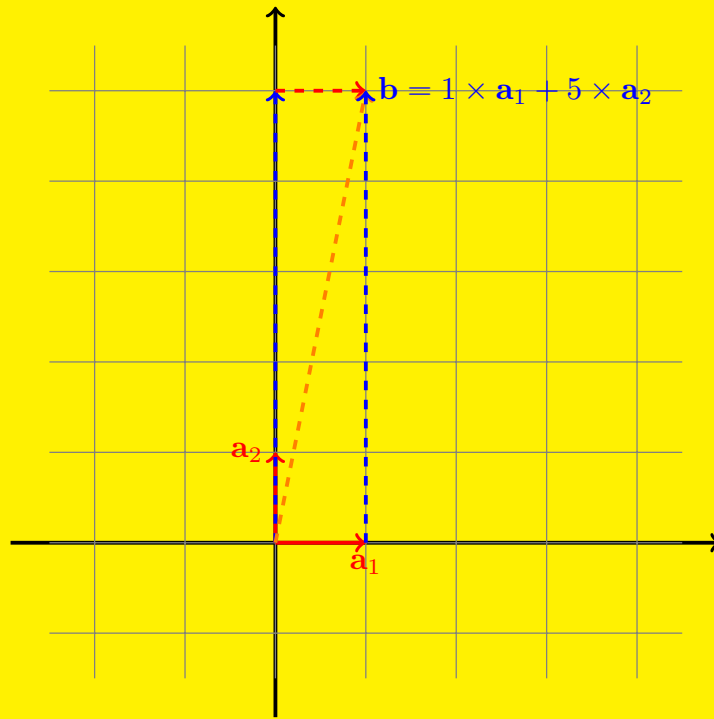
(c) The first column of $\mathbf{A}$ is all ones, so in order to get 1 we simply add 2 and subtract 1. Similarly, to get 5, subtract 1 from 6. In other words, try $x = -1$ of column 1 and $y = 1$ of column 2, and you get

$$-1 + (2) = 1$$

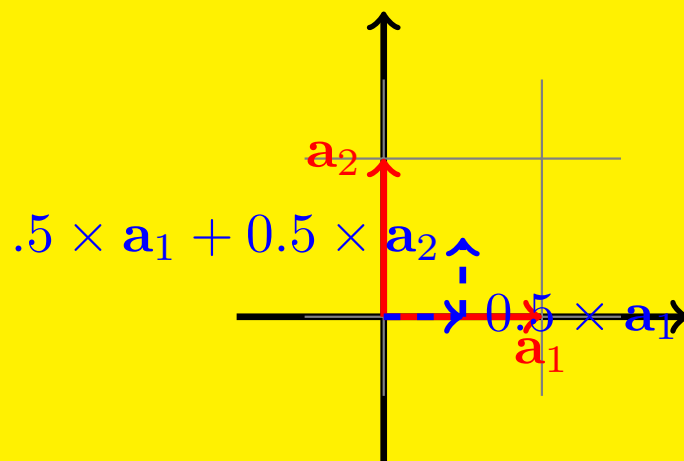
and

$$-1 + 6 = 5$$

(d) Here $\mathbf{A} = \mathbf{I}$, so we know that $\mathbf{x} = \mathbf{b}$, but it is still instructive to think of it as a linear combination of columns. For the first coordinate, we need one unit and for the second we need 5 units.



(e) Again, $\mathbf{A} = \mathbf{I}$, so we know that $\mathbf{x} = \mathbf{b}$. In the drawing though, we now only take half of $\mathbf{a}_1$



2. Linear combinations

Find $\mathbf{C}$

$$x \begin{bmatrix} 1 \\ 3 \end{bmatrix} + y \begin{bmatrix} 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{C}$$

In (1) you already used that

$$\mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix} = x\mathbf{a}_1 + y\mathbf{a}_2$$

hence $\mathbf{C} = \mathbf{0}_{2 \times 1}$

3. Using linear combinations in matrix multiplication

Find $\mathbf{B}$ in

$$\mathbf{AB} = \mathbf{C}$$

Use the fact that the columns of $\mathbf{C}$ are determined by equations like in (2). In other words, guess the columns of $\mathbf{B}$ one at a time.

(a)

$$\begin{bmatrix} 1 & 2 \\ 1 & 6 \end{bmatrix} \mathbf{B} = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 2 \\ 1 & 6 \end{bmatrix} \mathbf{B} = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$

(c)

$$\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \mathbf{B} = \begin{bmatrix} 3 & 6 \\ 3 & 6 \end{bmatrix}$$

(d)

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 3 & 3 \end{bmatrix} \mathbf{B} = \begin{bmatrix} 3 & 2 & 3 \\ 3 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

(a) In (1)(b) you found that

$$\begin{bmatrix} 1 & 2 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

In (1)(c) you found that

$$\begin{bmatrix} 1 & 2 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

Letting

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 6 \end{bmatrix}, \mathbf{x}_1 = \begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix}, \text{ and } \mathbf{x}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

We have that

$$\mathbf{C} = [\mathbf{Ax}_1 \quad \mathbf{Ax}_2]$$

hence

$$\mathbf{B} = [\mathbf{x}_1 \quad \mathbf{x}_2]$$

We can also write this directly as a combination of the columns of $\mathbf{A}$ by the rows of $\mathbf{B}$:

$$\mathbf{AB} = \left[\begin{bmatrix} 1 \times \begin{bmatrix} 1 \\ 1 \end{bmatrix} & + & \frac{1}{2} \times \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right] \left[\begin{bmatrix} (-1) \times \begin{bmatrix} 1 \\ 1 \end{bmatrix} & + & 1 \times \begin{bmatrix} 2 \\ 6 \end{bmatrix} \end{bmatrix} \right] = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$$

(b) Again, using the results in (1)(b) and (c), We have that

$$\mathbf{C} = [\mathbf{Ax}_2 \quad \mathbf{Ax}_1]$$

hence

$$\mathbf{B} = [\mathbf{x}_2 \quad \mathbf{x}_1]$$

(c) The matrix $\mathbf{C}$ is a multiple of $\mathbf{A}$. There are several solutions. One is that

$$\mathbf{C} = 3\mathbf{A}$$

which we can write as

$$\mathbf{C} = [\mathbf{Ax}_1 \quad \mathbf{Ax}_2]$$

where

$$\mathbf{x}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}, \text{ and } \mathbf{x}_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

hence

$$\mathbf{B} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

But you could also verify that a solution is, for example,

$$\mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 3 \end{bmatrix}$$

In questions (a) and (b) you can find $\mathbf{B}$ as $\mathbf{A}^{-1}\mathbf{C}$. In this question, question (c), you will note that $\mathbf{A}$ has *two identical rows* and hence $\mathbf{A}$ is *not invertible*

(d) Lets look at the matrix $\mathbf{C}$ column by column. First column

$$3 \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 0 \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 0 \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$$

Second column

$$0 \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 2 \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 0 \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

Third column

$$0 \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 0 \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 3 \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$$

To get this linear combination we have the diagonal matrix

$$\mathbf{B} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Similar to question (c), you will see that $\mathbf{A}$ has two columns that are identical and, hence, $\mathbf{A}$ is *not invertible*. You cannot find $\mathbf{B}$ by premultiplying $\mathbf{C}$ with the inverse of $\mathbf{A}$.

4. More linear combinations

Find $\mathbf{A}$ in

$$\mathbf{A} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} = \mathbf{I}$$

Hint, try to figure out what the first column of $\mathbf{A}$ might be (or simply figure out how you get each column of $\mathbf{I}$).

We see that $\mathbf{A}$ must satisfy the two equations

$$\mathbf{A} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

and

$$\mathbf{A} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

From the first, we see that $a_{21} = 0$ and $a_{11} = 1$. To get the second column of $\mathbf{A}$,

$$\mathbf{A} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = -2 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

So,

$$\begin{bmatrix} -2 \\ 0 \end{bmatrix} + \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

where we immediately see that $a_{12} = 2$ and $a_{22} = 1$, and

$$\mathbf{A} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

Congrats, you have just inverted a matrix